Paper 7 - Discrete-Continuous Bridge

CQE-paper-07
CQECMPLX Formal Paper Series
Review-facing PDF built 2026-06-12

Construction Basis

Define the bridge where discrete state changes approximate continuous dynamics through indexed windows.

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-07.md
- paper kernel manifest: production\paper-kernels\papers\CQE-paper-07\paper_kernel_manifest.json
- body: production\papers\CQE-paper-07\PAPER-BODY.md
- source: production\papers\CQE-paper-07\SOURCE.md
- formal: production\papers\CQE-paper-07\01-CQE-formal\FORMAL.md
- tool: production\papers\CQE-paper-07\02-CQE-tool\TOOL.md
- workbook: production\papers\CQE-paper-07\03-CQE-workbook\WORKBOOK.md
- recovered_output: production\lib-forge\recovered\papers_output\CQE-paper-07.md

Paper 7 - Discrete-Continuous Bridge

Abstract

Paper 7 proves the sample-preserving bridge used by the CQECMPLX suite. The previous papers produce discrete receipts: correction rows, registered coordinates, repair constraints, path states, and causal edges. Paper 7 proves that a finite indexed numeric trace can be embedded in a continuous piecewise-linear presentation without losing any original sample.

The theorem is exact and limited:

```
finite indexed trace -> continuous bridge
continuous bridge -> exact agreement at every indexed sample
```

It does not prove that the interpolant is the unique physical dynamics between samples. Between-sample dynamics require a separate theorem.

Claims

Claim 7.1. Every finite numeric discrete trace admits a continuous piecewise-linear interpolant.

Claim 7.2. The interpolant preserves every indexed sample exactly.

Claim 7.3. Adjacent segments agree at shared endpoints.

Claim 7.4. Sample preservation does not prove between-sample physical dynamics.

Definitions

A **discrete trace** is:

```
D = [(0,x0), (1,x1), ..., (n,xn)]
```

A **sample-preserving bridge** is a continuous function `F` on `[0,n]` such that:

```
F(k) = xk
```

for every integer sample index `k`.

For `t in [k,k+1]`, define the piecewise-linear bridge:

```
a = t - k
F(t) = (1-a) * xk + a * x(k+1)
```

Theorem 7.1 - Sample-Preserving Bridge

Every finite discrete trace over numeric values admits a continuous piecewise-linear bridge that exactly preserves all indexed samples.

Proof

Between each adjacent pair `(k,xk)` and `(k+1,x(k+1))`, draw the straight segment joining the two values:

```
F(t) = (1-a) * xk + a * x(k+1)
```

where `a = t-k`. At the left endpoint, `a=0`, so `F(k)=xk`. At the right endpoint, `a=1`, so `F(k+1)=x(k+1)`. Adjacent segments therefore agree at the shared integer sample, and the full piecewise function is continuous.

Since each integer sample is an endpoint of one or two adjacent segments and all such segments agree at that endpoint, every original sample is preserved exactly.

Receipt

The primary executable receipt is:

```
production/formal-papers/CQE-paper-07/verify_discrete_continuous_bridge.py
production/formal-papers/CQE-paper-07/discrete_continuous_bridge_receipt.json
```

The receipt status is `pass`. It verifies:

```
integer_samples_preserved                  = true
max_sample_error_is_zero                   = true
adjacent_segments_share_endpoints          = true
rule30_rule90_correction_identity_holds    = true
between_sample_physics_left_as_obligation = true
```

The receipt trace is:

```
[0,1,1,0,1,0,0,1]
```

All sample errors are zero.

Falsifier

The verifier rejects this claim:

```
sample-preserving interpolation proves all between-sample physics
```

That claim is false. Sample preservation proves exact agreement at indexed samples only.

Relation to Earlier Papers

Paper 6 gives typed causal edges. Paper 7 gives a continuous presentation of indexed proof data while preserving the discrete receipt.

Paper 2's correction identity remains discrete:

```
Rule30(L,C,R) = Rule90(L,R) xor (C and not R)
```

Paper 7 may draw those values as a continuous curve, but the proof remains at the indexed samples unless another theorem closes the between-sample dynamics.

Role in the Suite

The bridge rule is:

```
discrete receipt -> indexed trace -> continuous presentation
```

The forbidden move is:

```
continuous presentation -> erased discrete receipt
```

This lets later papers use continuous-looking fields without losing the original finite proof.

Open Obligations

- Expose `verify_discrete_continuous_bridge` through the installable
kernel/API interface.
- Add separate theorems for Hamiltonian or physical dynamics between samples.
- Preserve causal receipt links whenever interpolated traces are used.

- Carry bridge residuals into later continuum papers only as obligations until verified.

Conclusion

Paper 7 proves a safe bridge from discrete receipts to continuous presentation. It preserves every indexed sample exactly and refuses to convert visual smoothness into unsupported physical dynamics.